

TEL-HAI
University of Kiryat Shmona
& the Galilee

Machine Learning and Data Analysis

Rule-Based Classification & Model Evaluation

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Lecture Agenda

This lecture walks you through the Titanic example from data exploration to rule-based model evaluation. By the end of this session, you will understand how raw data is transformed into rule-based predictions and evaluated using standard metrics.

Titanic: History & Context

From the real ship to the movie, forming a hypothesis about survival

Data Collection & Exploration

Dataset features, first look, feature types, and EDA visualizations

Rule-Based Classification

Applying "Women & Children First" rules, implementing in Python, Jack vs. Rose profiles

Classification Evaluation Metrics

Confusion matrix, Accuracy (78.7%), Recall (68.1%), Precision (74.2%), and F1 Score — evaluating our rule-based classifier

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Reminder of What We Did in Previous Lab Sessions

✅ Problem Formulation

Define the question clearly before touching any data.

✅ Data Collection

Gather relevant data from databases, APIs, surveys, or sensors.

✅ Cleaning & Preprocessing

Handle missing values, remove noise, and fix inconsistencies.

📍 Exploratory Data Analysis

Visualize distributions, find correlations, spot anomalies.

Current Step

Modeling & Evaluation

Build predictive models, validate with metrics.

Deployment & Monitoring

Put models into production, track performance over time.

Results Dissemination

Present findings clearly to stakeholders through reports, dashboards, and storytelling.

TITANIC EXAMPLE

The Titanic — A Brief History

Reminder of What We Did in Previous Lab Sessions

💀 1,500+

Passengers and crew who perished in the disaster

🚢 2,224

Total people aboard the ship on its ill-fated voyage

📅 April 15, 1912

Date the Titanic sank after striking an iceberg

The RMS Titanic was the largest and most luxurious ocean liner of its time. On its maiden voyage from Southampton to New York City, it struck an iceberg in the North Atlantic at 11:40 PM on April 14, 1912, and sank in the early hours of April 15, 1912.

> This tragedy became a landmark case study in survival analysis, making it the perfect foundation for learning machine learning concepts.



FILM REFERENCE

Titanic — The Movie

James Cameron's 1997 blockbuster *Titanic* brought the disaster to life through the fictional love story of Jack Dawson and Rose DeWitt Bukater. The film grossed over **$2.2 billion** at the box office and won 11 Academy Awards, including Best Picture.

🎬 Watch the official trailer: https://youtu.be/kVrqfYjkTdQ — As you watch, think about the characters' backgrounds and how they might relate to survival probability based on what the data tells us.

Throughout this module, we will use Jack and Rose as real-world test cases to validate what our exploratory data analysis reveals about survival patterns on the Titanic.





PLANNING — RESEARCH HYPOTHESIS

Titanic Example: Forming a Hypothesis

Before writing a single line of code, every ML project begins with a clear research question. In the Titanic case, we start with the most important question:

> **"Given the personal information of a passenger aboard the Titanic, can we predict whether they survived or perished?"**

The Input

Personal attributes of each passenger — age, sex, ticket class, fare, family size, and port of embarkation

The Output (Target)

A binary prediction: **Survived (1)** or **Did Not Survive (0)**

The Fun Question

Based on the data, is the ending of the 1997 film *exactly* what we would expect? 🎬

PLANNING — DATA COLLECTION

The Titanic Dataset

The Titanic dataset is one of the most widely used benchmark datasets in machine learning education. It is an ideal starting point because it is small enough to understand fully, yet rich enough to demonstrate real analytical challenges.

891

Passengers

Total records in the dataset, each representing one person aboard the RMS Titanic

12

Features

Variables available per passenger, ranging from demographics to ticket information

1

Target Variable

Binary outcome — **Survived** (1) or **Did Not Survive** (0)

The dataset is publicly available and maintained on platforms like **Kaggle**, where it serves as the entry point for countless machine learning competitions and tutorials.

PLANNING — DATA COLLECTION

Titanic Dataset: Feature Description

Understanding what each column represents is a critical first step. Below are the 12 features available in the dataset:

Key Predictive Features

- **Pclass** — Socioeconomic status proxy
- **Sex** — Binary gender indicator
- **Age** — Continuous passenger age
- **Fare** — Ticket price paid
- **SibSp / Parch** — Family size indicators
- **Embarked** — Port of boarding

Low-Value / Useless Features

- **PassengerId** — Unique ID, no predictive value
- **Name** — Text identifier
- **Ticket** — Alphanumeric ticket number
- **Cabin** — Mostly missing values

⚠ Features with no predictive signal or too many missing values are typically dropped during preprocessing.

PLANNING — DATA COLLECTION

Titanic Data — A First Look

Before any analysis, we always inspect the raw data. This step, often called **df.head()** in pandas, gives us a feel for what we're working with — the shape of the data, data types, and any obvious anomalies.

A	B	C	D	E	F	G	H	I	J	K	L
PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	0	3	Saundercock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson, Mr. Anders Johan	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55	0	0	248706	16		S
17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125		Q
18	1	2	Williams, Mr. Charles Eugene	male		0	0	244373	13		S
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31	1	0	345763	18		S

Mixed Data Types

The dataset contains numeric, categorical, and text columns — each requiring different preprocessing strategies.

Missing Values

Columns like **Age** and **Cabin** contain blank entries that must be handled before modeling can begin.

Target Column

The **Survived** column (0 or 1) is our label — what we are trying to teach the model to predict.

PLANNING — DATA COLLECTION

Titanic Dataset: Feature Types

Classifying features by their data type is essential — it determines which preprocessing steps and algorithms are appropriate. Here are the six key feature types found in the Titanic dataset:



Categorical / Nominal

Multiple categories with no inherent order. **Example:** Embarked (C, Q, S)



Binary

Only two possible values. **Example:** Sex (Male / Female)



Ordinal

Categorical with a meaningful order. **Example:** Pclass (1, 2, 3)



Continuous

Any value in a range. **Examples:** Age, Fare



Count

Integer counts of something. **Examples:** SibSp, Parch



Useless

No predictive signal. **Examples:** PassengerId, Name, Cabin, Ticket

PRE-PROCESSING — DATA EXPLORATION

Data Exploration: Understanding Your Data

Data exploration is the first step in pre-processing. It helps reveal patterns, anomalies, and early insights before modeling.



Characterize Data Distribution

Check the spread and shape of each variable.



Identify Relationships & Structure

Look for correlations and dependencies between variables.



Validate Correctness & Relevance

Confirm the data is accurate and useful for the task.

PRE-PROCESSING — DATA EXPLORATION

Data Exploration: Visualizing Relationships

We explore how each feature relates to the target, **Survived**, to spot useful predictors and patterns early.

Categorical & Binary Features

Use bar or stacked bar charts for **Sex**, **Pclass**, and **Embarked**.

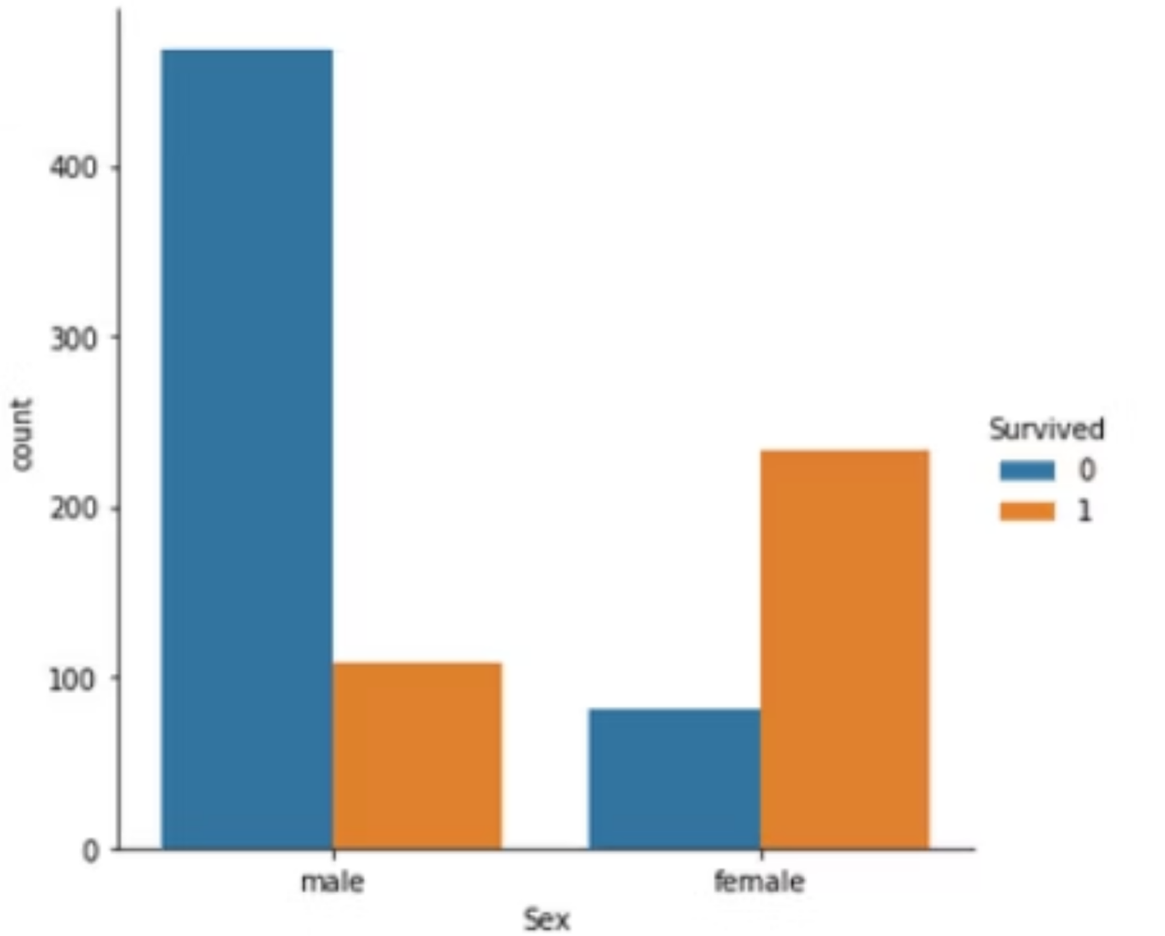
Continuous & Count Features

Use histograms, box plots, or violin plots for **Age**, **Fare**, **SibSp**, and **Parch**.

These views quickly reveal correlations and guide feature engineering.

PRE-PROCESSING — DATA EXPLORATION

Survival by Sex



Key Finding

Gender was one of the strongest predictors of survival on the Titanic.

♂ ~20%

Male survival rate

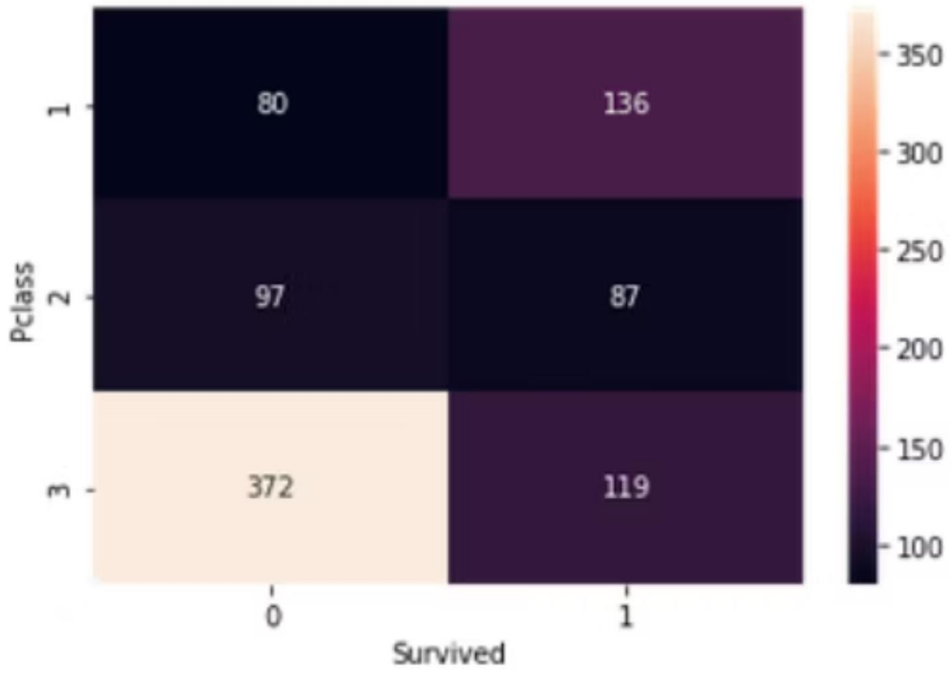
♀ ~75%

Female survival rate

The "women and children first" maritime protocol during evacuation heavily influenced who made it onto the lifeboats.

PRE-PROCESSING — DATA EXPLORATION

Survival by Passenger Class



What the Heatmap Shows

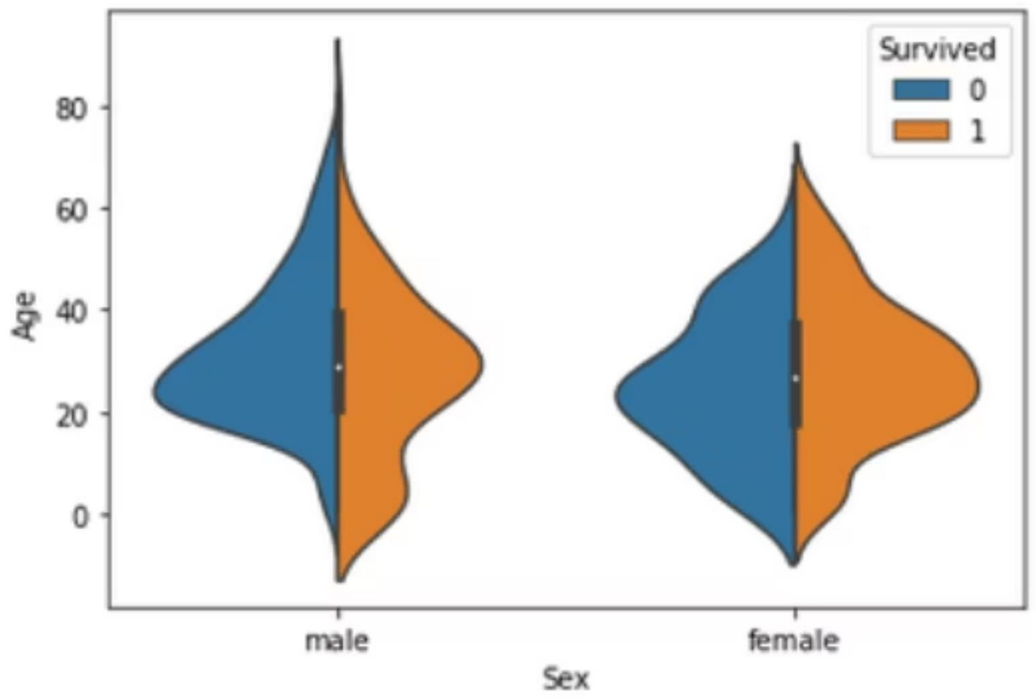
The heatmap cross-tabulates **Pclass** against **Survived**. Lighter cells indicate higher passenger counts.

- **Class 1:** 136 survived vs. 80 perished — survival *majority*
- **Class 2:** 87 survived vs. 97 perished — roughly balanced
- **Class 3:** Only 119 survived vs. 372 perished — clear disadvantage

⚠ Why? First-class cabins were located on upper decks, closer to the lifeboats. Wealth and social status directly influenced access to safety.

PRE-PROCESSING — DATA EXPLORATION

Survival by Age and Sex



Reading the Violin Plot

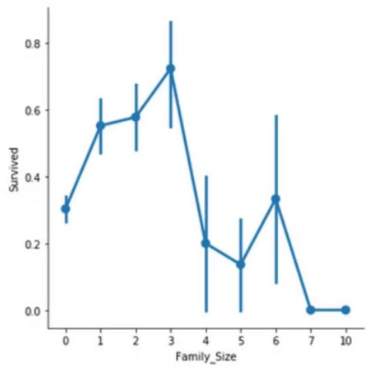
The width of each violin at a given age reflects the density of passengers at that age. Orange = Survived, Blue = Did Not Survive.

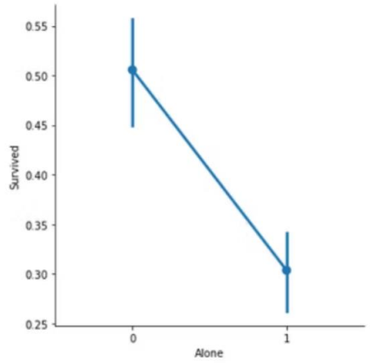
- ✅ **Children** (both sexes): strong survival signal
- ✅ **Women aged 20–50:** high survival rate
- ❌ **Men overall:** survival drops significantly with age

Age interacts with sex to create a nuanced survival pattern — a key insight for feature engineering.

PRE-PROCESSING — DATA EXPLORATION

Survival by Family Size





Engineered Features

Two new features were derived from existing columns:

- `Family_Size = Parch + SibSp` — total family members aboard
- `Alone = 1 if Family_Size == 0` — traveling solo flag

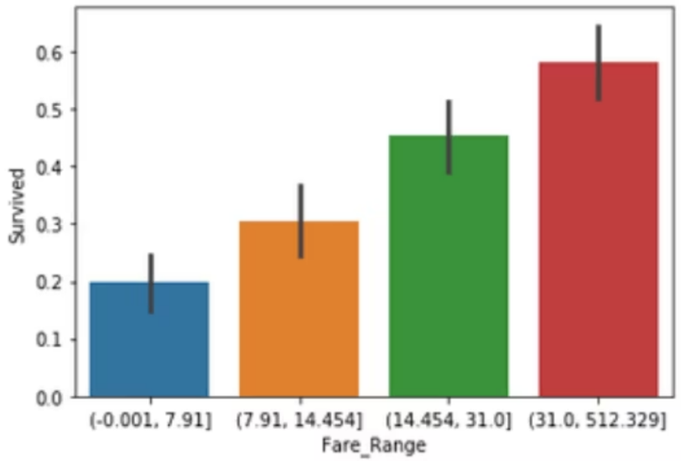
Key Insights

- Passengers traveling **alone** had a significantly lower survival rate
- Small families (2–4 members) fared the best
- Very large families (**5+ members**) had poor survival — harder to evacuate together

Feature engineering — creating new variables from existing ones — is a powerful technique to surface hidden patterns.

PRE-PROCESSING — DATA EXPLORATION

Survival by Fare Paid



The Finding

Passengers who paid a **higher fare** had a significantly greater chance of survival. This feature is closely correlated with **Pclass** — wealthier passengers bought first-class tickets and boarded in better deck positions.

Why Does This Matter?

When two features are highly correlated (like Fare and Pclass), we must be careful of **multicollinearity** in certain models. However, for tree-based models, both features can still contribute independently.

⚠ Always check for feature correlation before building your model — redundant features can sometimes harm performance.

PRE-PROCESSING — DATA EXPLORATION

Feature Correlation: Pearson Correlation

Pearson correlation measures the linear relationship between two continuous variables, ranging from -1 to +1.

$$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$$

Numerator — Do They Move Together?

Measures how much x and y deviate from their means at the same time. If both go up together (or down together), the product is positive. If they move in opposite directions, it's negative.

Denominator — Normalize the Scale

Scales the result by the total spread of each variable independently. This ensures r always stays between -1 and +1, regardless of the units or magnitude of x and y.

+1

Perfect Positive Correlation

0

No Linear Relationship

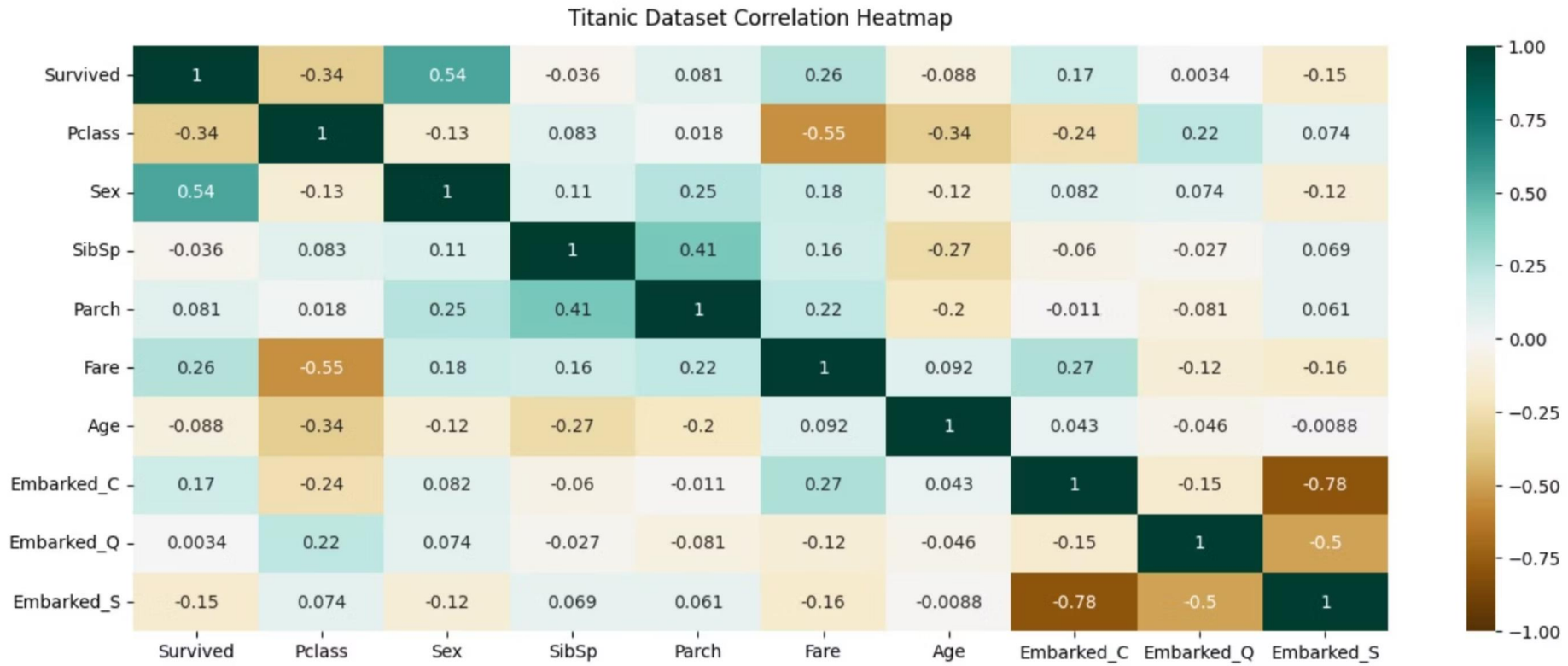
-1

Perfect Negative Correlation

PRE-PROCESSING — DATA EXPLORATION

Titanic Dataset Correlation Heatmap

This heatmap shows Pearson correlations across the Titanic features.



Two key patterns stand out: Sex → Survived (0.54) and Pclass → Fare (-0.55).

These relationships highlight the strongest signals in the dataset.



PRE-PROCESSING — DATA EXPLORATION

Is "The End" as Expected?

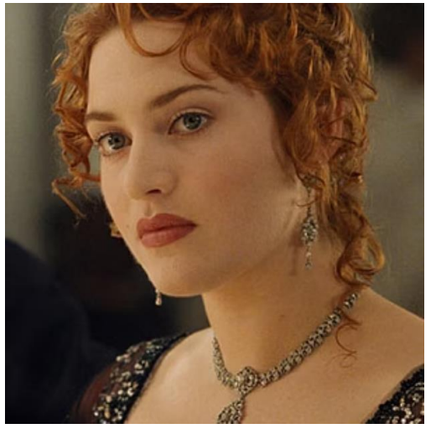
Now that we have explored the data, let's put our findings to the test. Before we build any formal model, our exploratory data analysis has already revealed powerful patterns: **gender, class, age, fare, and family size** all significantly influence survival outcomes.

🎬 Watch the film's ending scene: https://youtu.be/mct-F_bp1Qw?t=31

As you watch, ask yourself: based on everything the data has told us, is the outcome of Jack and Rose's story *exactly* what a trained ML model would predict? The answer might surprise you — or confirm exactly what you already suspect.

Passenger Profiles: Jack vs. Rose

Let's apply our EDA findings to the two main characters. Based on their personal profiles, what does the data suggest about each passenger's likelihood of survival?



Rose DeWitt Bukater

- ♀ **Woman**
- 🎂 Young (early 20s)
- 💰 Wealthy — 1st Class
- 👨‍👩‍👧 Traveling with family

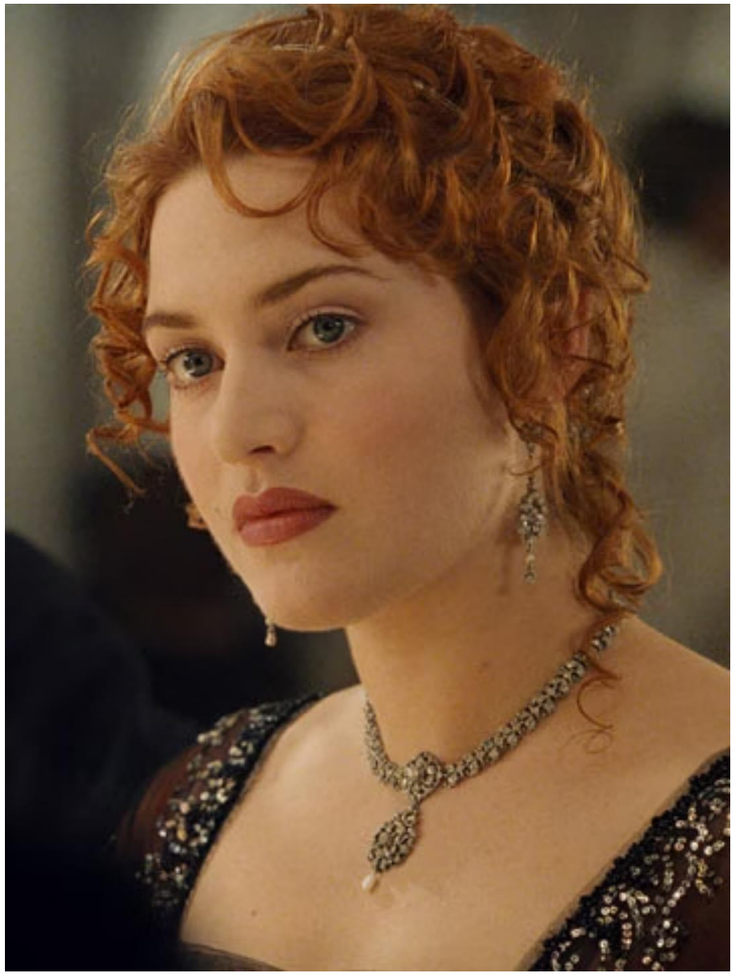


Jack Dawson

- ♂ **Man**
- 🎂 Young (early 20s)
- 💸 Poor — 3rd Class
- 🧍 Traveling alone

Passenger Survival Analysis from EDA

Applying the insights from our exploratory data analysis to Jack and Rose, the outcome is clear — and statistically predictable. Each survival factor we identified in the data points strongly in one direction.





Rose — High Survival Probability ✅

- ♀ Female → ▯ ↑ High survival rate for women
- 🎩 1st Class → ▯ ↑ Upper deck, lifeboat access
- 👶 Young age → ▯ → Weak signal (not decisive)
- 👨‍👩‍👧 Traveling with family → ▯ ↑ Not alone

Jack — Low Survival Probability ❌

- ♂ Male → 🔴 ↓ ~20% base rate
- 🧍 3rd Class → 🔴 ↓ Lower decks, restricted access
- 💸 Low fare → 🔴 ↓ Negative signal
- 👶 Young age → ▯ → Weak signal (not decisive)
- 🧍 Traveling alone → 🔴 ↓ Further reduces odds

We could predict the ending with high certainty! 🎬

PRE-PROCESSING — DATA EXPLORATION

Rule-Based Survival Prediction

Our exploratory analysis highlighted a strong predictor. We can formalize this insight into a single, explicit rule to create a simple, interpretable baseline for survival prediction.

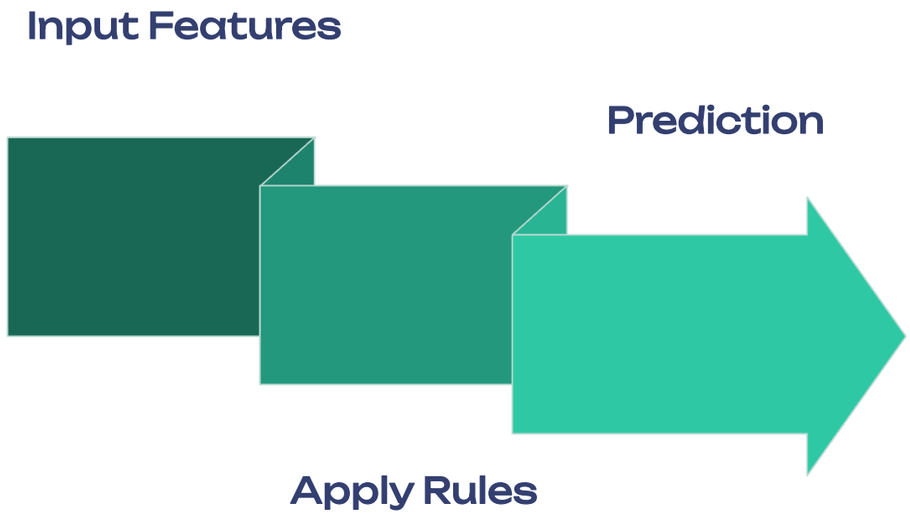
Rule: Gender-Based Survival

IF (**Sex** is Female) THEN Survived ELSE Perished

Rule: Women & Children First

IF (**Sex** is Female) OR (**Age** < 12) THEN Survived ELSE Perished

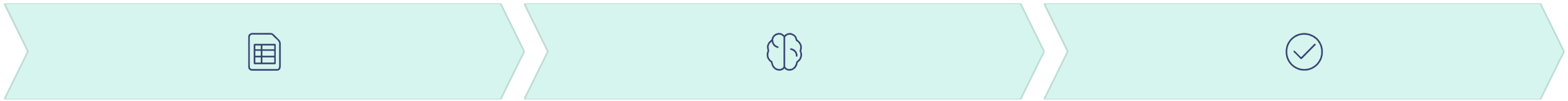
These rules, directly derived from our EDA findings, demonstrate how initial data exploration can yield powerful, understandable models even before applying complex machine learning algorithms.



MACHINE LEARNING FUNDAMENTALS

Survival Prediction: Rule-Based Classification

Having explored the data and identified key patterns, we now use explicit, hand-crafted rules derived from EDA — not a learned model — to classify passengers as survived or perished.



Input Features

Passenger data: Sex, Age, Class, Fare, Family Size

Rule-Based Classifier

Explicit IF/ELSE rules derived from EDA insights.

Predicted Outcome

Categorical label: Survived (1) or Perished (0)

Before machine learning became widespread, rule-based classification was the dominant approach. Experts manually encoded domain knowledge into explicit rules, making the method intuitive and interpretable. But it does not scale well: every rule has to be written by hand, and the system cannot adapt to new patterns in data. This limitation helped motivate machine learning, where algorithms learn rules automatically from data.

MACHINE LEARNING FUNDAMENTALS

Gender-Based Rule: How It Works

The simplest rule uses a single feature — Sex — to predict survival. If a passenger is female, we predict Survived = True. Otherwise, Perished = False.

PassengerId	Survived	Pclass	Name		Age	
1	0	3	Mr. Owen Harris Braund	male	22.0	❌ False
2	1	1	Mrs. John Bradley Cumings	female	38.0	✅ True
3	1	3	Miss. Laina Heikkinen	female	26.0	✅ True
4	1	1	Mrs. Jacques Heath Futrelle	female	35.0	✅ True
5	0	3	Mr. William Henry Allen	male	35.0	❌ False

♀ female → ✅ Survived (True)

♂ male → ❌ Perished (False)

MACHINE LEARNING FUNDAMENTALS

Implementing Rule-Based Prediction in Python

We can apply our "Women & Children First" rule directly to the Titanic dataset using Python to generate a new prediction column. This new column, often called `Prediction` or `Predicted_Survival`, will be based purely on our simple rules and will serve as a baseline for evaluation.

```
# Initialize a new column 'Prediction' with default value 0 (Perished)
df['Prediction'] = 0

# Apply the "Women First" rule: Set Prediction to 1 for all females
df.loc[df['Sex'] == 'female', 'Prediction'] = 1

# Apply the "Children First" rule: Set Prediction to 1 for children (Age < 12)
# This rule will override for male children, as per "Women & Children First"
df.loc[df['Age'] < 12, 'Prediction'] = 1

# Display the first few rows with Sex, Age, Survived, and Prediction columns
print(df[['Sex', 'Age', 'Survived', 'Prediction']].head())
```

This programmatic implementation creates a tangible output: a `Prediction` column that mirrors the logic of our rule. We can then directly compare these predictions against the actual `Survived` column to assess the rule's effectiveness and identify areas where it succeeds or fails. This forms the basis for model evaluation.

MACHINE LEARNING FUNDAMENTALS

Evaluating Rule Performance: The Confusion Matrix

After making predictions with our rule, the next critical step is to evaluate how well it performed. The **Confusion Matrix** is a fundamental tool for assessing the accuracy of a classification model, providing a detailed breakdown of correct and incorrect predictions.

	Predicted Negative (Perished)	Predicted Positive (Survived)
Actual Negative (Perished)	**468** True Negative ✅	**81** False Positive ❌
Actual Positive (Survived)	**109** False Negative ❌	**233** True Positive ✅

This matrix categorizes each prediction into one of four outcomes, helping us understand where our model is succeeding and failing.

True Positives (TP)

Correctly predicted positives. E.g., The model predicted survival, and the passenger indeed survived.

True Negatives (TN)

Correctly predicted negatives. E.g., The model predicted perishing, and the passenger indeed perished.

False Positives (FP)

Incorrectly predicted positives (Type I error). E.g., The model predicted survival, but the passenger perished.

False Negatives (FN)

Incorrectly predicted negatives (Type II error). E.g., The model predicted perishing, but the passenger survived.

MACHINE LEARNING FUNDAMENTALS

Performance Metrics: Accuracy

Accuracy is often the first metric people consider, as it's straightforward: it measures the proportion of total predictions that were correct. While intuitive, it doesn't always tell the whole story, especially with imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

	Predicted Negative	Predicted Positive
Actual Negative	**468** TN ✅	**81** FP ❌
Actual Positive	**109** FN ❌	**233** TP ✅

Accuracy uses the highlighted diagonal cells (TN + TP = 468 + 233 = 701 correct predictions out of 891 total).

Plugging these values into the formula:

$$Accuracy = \frac{233 + 468}{233 + 468 + 81 + 109} = \frac{701}{891} \approx 0.787$$

This means our simple gender-based rule achieved an accuracy of approximately **78.7%** on the Titanic dataset, correctly classifying nearly 8 out of 10 passengers.

MACHINE LEARNING FUNDAMENTALS

Performance Metrics: Recall

While Accuracy tells us how many predictions were correct overall, **Recall** (also known as Sensitivity or True Positive Rate) focuses specifically on the positive class. It measures how many of the actual positive cases our model correctly identified. It's crucial when the cost of missing a positive case (False Negative) is high.

$$Recall = \frac{TP}{TP + FN}$$

	Predicted Negative	Predicted Positive
Actual Negative	**468** TN	**81** FP
Actual Positive ← Recall	**109** FN ❌	**233** TP ✅

Recall specifically uses the highlighted cells from the confusion matrix that represent actual positive cases: True Positives (TP) and False Negatives (FN). These are the passengers who actually survived.

Plugging these values into the formula:

$$Recall = \frac{233}{233 + 109} = \frac{233}{342} \approx 0.681$$

Our gender-based rule has a Recall of approximately **68.1%**. This means that out of all passengers who actually survived, our rule correctly identified about 68.1% of them. This is a measure of the model's ability to "find" all positive samples.

MACHINE LEARNING FUNDAMENTALS

Performance Metrics: Precision

While Recall measures how many actual positive cases our model found, **Precision** (also known as Positive Predictive Value) focuses on the accuracy of the positive predictions made. It answers: "Of all the instances our model predicted as positive, how many were actually correct?" It's crucial when minimizing false positives is a priority, such as in spam detection or medical diagnoses.

$$Precision = \frac{TP}{TP + FP}$$

	Predicted Negative	Predicted Positive ← Precision
Actual Negative	**468**	**81**
	TN	FP ❌
Actual Positive	**109**	**233**
	FN	TP ✅

Precision specifically uses the highlighted cells from the confusion matrix that represent cases predicted as positive: True Positives (TP) and False Positives (FP). These are the passengers our rule predicted would survive.

Plugging these values into the formula:

$$Precision = \frac{233}{233 + 81} = \frac{233}{314} \approx 0.742$$

Our gender-based rule has a Precision of approximately **74.2%**. This means that out of all passengers our rule predicted would survive, about 74.2% actually did. This metric indicates the model's ability to avoid false alarms and make reliable positive predictions.

MACHINE LEARNING FUNDAMENTALS

Performance Metrics: F1 Score

The **F1 Score** is a harmonic mean of Precision and Recall, offering a single metric that balances both. It's particularly useful when you need to seek a balance between false positives and false negatives, especially in cases where there's an uneven class distribution.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

	Predicted Negative	Predicted Positive
Actual Negative	**468** TN	**81** FP ❌
Actual Positive	**109** FN ❌	**233** TP ✅

The F1 Score combines the values of True Positives (TP), False Positives (FP), and False Negatives (FN) to provide a comprehensive view of the model's performance on the positive class.

Using the Precision (~0.742) and Recall (~0.681) calculated earlier:

$$F1\ Score = 2 \times \frac{0.742 \times 0.681}{0.742 + 0.681} = 2 \times \frac{0.5053}{1.423} \approx 0.710$$

Our gender-based rule has an F1 Score of approximately **71.0%**. This indicates a fair balance between correctly identifying actual survivors and avoiding false predictions of survival. The F1 score helps in comparing models, especially when one might have high precision but low recall, or vice-versa.



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& the Galilee

Thank You!

We appreciate your time and interest in understanding the nuances of data cleaning. We look forward to collaborating on your data science initiatives.

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Reception Hours

By Appointment Only